MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – C)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 25**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. This question paper contains 11 compulsory questions divided into four Sections: A, B, C, and D.
2. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each (Q1 to Q5).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each (Q6 and Q7).
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each (Q8 to Q10).
5. **Section D** comprises 1 Long Answer (LA) question of 5 marks (Q11). An internal choice is provided.
6. Draw neat, labeled geometric figures wherever applicable. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The length of the shadow of a tower on the plane ground is $\sqrt{3}$ times the height of the tower. The angle of elevation of the Sun is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

Q2. Raina is 1.5 m tall. At an instant, his shadow is 1.8 m long. At the same time, the shadow of a tree is 9 m long. The height of the tree is: [1]

(A) 3 m (B) 4.5 m (C) 7.5 m (D) 10.8 m

Q3. A peacock sitting on the top of a tree of height 10 m observes a snake moving on the ground. If the snake is $10\sqrt{3}$ m away from the base of the tree, then the angle of depression of the snake from the eye of the peacock is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

Q4. From a point on the ground, which is 60 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be $30°$. The height of the tower is: [1]

(A) $20\sqrt{3}$ m (B) $30\sqrt{3}$ m (C) $60\sqrt{3}$ m (D) 20 m

Q5. **Assertion (A):** A ladder leaning against a wall stands at a horizontal distance of 6 m from the wall. If the height of the wall up to which the ladder reaches is 8 m, then the length of the ladder is 10 m.
Reason (R): The ladder makes an angle of $60°$ with the ground. [1]

(A) Both (A) and (R) are true and (R) is correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. A vertical pole of height 10 m casts a shadow of 15 m on the ground and at the same time, a tower casts a shadow of 45 m on the ground. Find the height of the tower. [2]

Q7. A 1.6 m tall boy stands at a distance of 3 m from a lamp-post and casts a shadow of length 4 m on the ground. Find the height of the lamp-post. [2]

SECTION C: Short Answer Questions (3 × 3 = 9 Marks)

Q8. Two boats are sailing in the sea 80 m apart from each other towards a cliff AB. The angles of depression of the boats from the top of the cliff are 30° and 45° respectively. Find the height of the cliff. (Take $\sqrt{3} = 1.732$) [3]

Q9. An aeroplane at an altitude of 200 metres observes the angles of depression of opposite points on the two banks of a river to be 45° and 60°. Find the width of the river. (Use $\sqrt{3} = 1.732$) [3]

Q10. From the top of a cliff 25 m high, the angle of elevation of the top of a tower is found to be equal to the angle of depression of the foot of the tower. Find the height of the tower. [3]

SECTION D: Long Answer Question (1 × 5 = 5 Marks)

Q11. The angle of elevation of the top of a tower 24 m high from the foot of another tower in the same plane is 60°. The angle of elevation of the top of the second tower from the foot of the first tower is 30°. Find the distance between the two towers and the height of the other tower. Also, find the length of the wire attached to the tops of both the towers. [5]

— OR —

From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30°. Determine the height of the tower. [5]

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